

ISA 634: Optimization Assignment 1

Part 1 – Review of Basics

Instructions: If you finish the lab outside of class time, please slide it under my office door.

1. (1 point) Give an example of a real-life decision that can be modeled as:
 - a. continuous variable
 - b. integer variable
 - c. binary variable
2. (1 point) A model is intended to help us capture and understand some (but not all) of the characteristics of the object or decision being examined.
 - a. True
 - b. False
3. (1 point) Pick the answer that best matches the following statement. In an optimization problem, a _____ is used to evaluate the quality/value of a given solution.
 - a. model
 - b. objective function
 - c. constraint
 - d. decision variable
4. (1 point) Pick the answer that best matches the following statement. In an optimization problem, a _____ represents limitations on the how we can make decisions.
 - a. model
 - b. objective function
 - c. constraint
 - d. decision variable
5. (1 point) The primary goal of any optimization is to try to identify decision variables values (i.e., a solution) that gives the best possible value (whether optimal or not) of the objective function.
 - a. True
 - b. False

6. (4 points) Find the **A** matrix, **c**^T vector and **b** vector for the optimization problem below:

$$\begin{array}{ll}\max & x_1 - 4x_3 \\ \text{s.t.} & x_1 + x_2 + 5x_3 - 8 \leq 0 \\ & x_1 - 2x_3 + 3 = 0 \\ & -3x_2 + 4x_3 - 3 \geq 10 \\ & x_1, x_2, x_3 \geq 0\end{array}$$

7. (5 points) Rewrite the model below in standard form where the objective is “min”, all constraints have a direction of “ $\leq$ ”, and all variables are nonnegative.

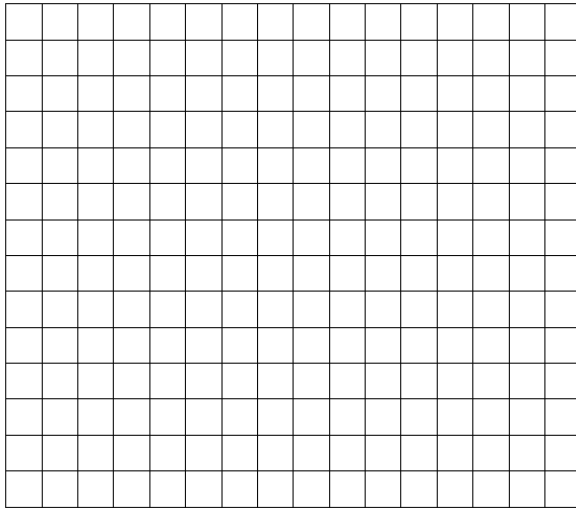
$$\begin{array}{ll}\max & -x_1 + 2x_2 \\ \text{s.t.} & x_1 + x_2 - 8 \leq 0 \\ & 3x_1 - 2x_2 = 5 \\ & -3x_2 \geq 10 \\ & x_1 \geq 0 \\ & x_2 \leq 0\end{array}$$

8. (2 points) Sketch the feasible region for the given set of constraints below:

$$x_1 \leq 4$$

$$x_1 \geq 0.5$$

$$x_1 \geq 1$$



9. (4 points) Sketch the feasible region for the given set of constraints below:

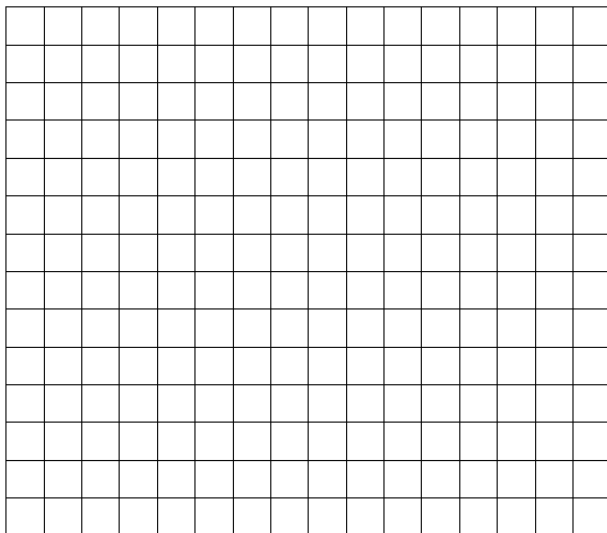
$$2x_1 + 5x_2 \leq 10$$

$$x_1 \leq 6$$

$$x_2 \leq 4$$

$$x_1 \geq 0$$

$$x_2 \geq 0$$



10. (4 points) Sketch the feasible region for the given set of constraints below:

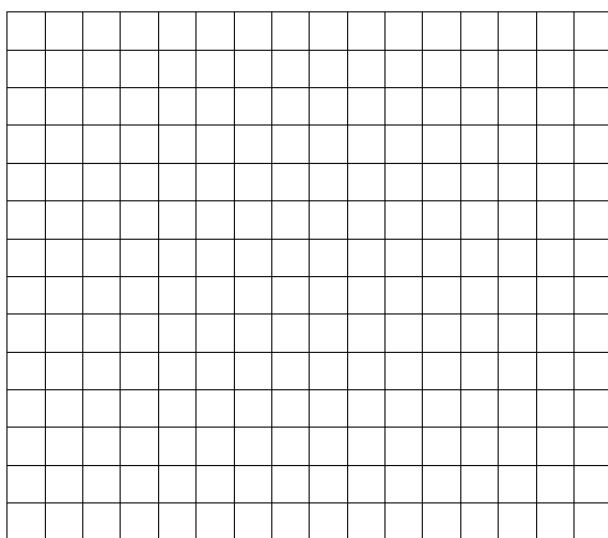
$$2x_1 + 4x_2 = 12$$

$$x_1 \leq 4$$

$$x_2 \leq 4$$

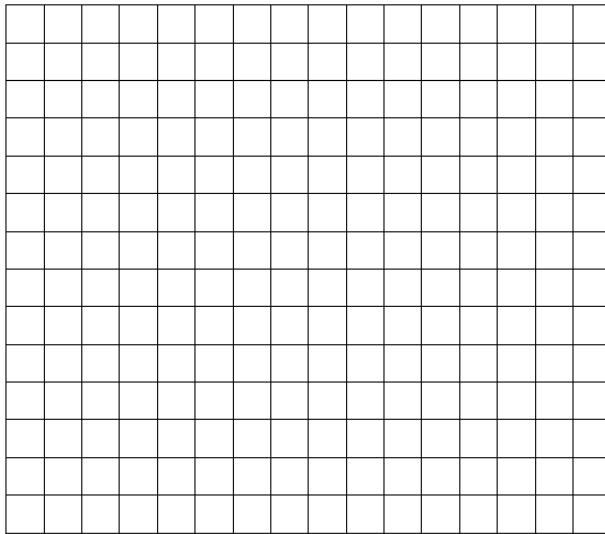
$$x_1 \geq 0$$

$$x_2 \geq 0$$



11. (10 points) Solve the model below using the graphical solution method (show feasible region, at least two iso-profit lines – including the optimal one, optimal solution, optimal objective value):

$$\begin{array}{ll}\min & x_1 + 2x_2 \\ \text{st} & -x_1 + x_2 \geq 2 \\ & x_1 \geq 0 \\ & x_2 \geq 0\end{array}$$



12. Suppose an investment newsletter firm is introducing two new products: 1) a weekly market report; and 2) a weekly trade strategy alert. Both products will be specifically customized to each customer's investment strategy and risk "appetite". The expected profit from the market report is \$59.99 per report and \$29.99 per trade strategy alert. The firm's management wants to establish a sales quota for the new products that maximizes the total expected profit. The work requirements associated with the two new products are summarized in the table below:

Department	Work-Hours per Unit		Work-hours Available
	Market report	Trade strategy alert	
Technical Analysis	5	3.5	2400
Writing & editorial	1	1	1225
Administration	1	0.5	710

Additionally, suppose that this is taking place before the internet, and everything had to be printed on paper and quickly shipped to their customers. Hypothetically, suppose the firm encountered a situation where they had a limited amount of paper due to some uncontrollable supply chain disruption. Namely, they have 7,000 sheets of paper in inventory. Assume that each market report uses 50 sheets of paper and each trade strategy alert uses 20 sheets of paper.

- a) (8 points) Formulate (do not solve) the linear programming model for this problem. Be sure to provide (in words) the meaning of the decision variables.
- b) (4 points) Demonstrate that the units of the objective function and each constraint are correct.

13. Solve the below linear programming model using Excel (please upload your spreadsheet onto Canvas).

$$\begin{array}{ll}\min & 1.3x_1 + 2.5x_2 + x_3 + 0.5x_4 + 3x_5 \\ \text{s.t.} & -2x_1 + 4x_2 + 5x_3 + x_4 + 4x_5 \leq 42 \\ & 3x_1 + x_2 + 2x_3 - 2x_4 + 7x_5 \leq 47 \\ & 7x_1 + 2x_2 + 3x_3 + 5x_4 + 3x_5 \leq 46 \\ & 5x_1 + x_2 + 3x_3 + 4x_5 \geq 28 \\ & 6x_1 - 3x_2 + x_3 + 4x_4 + 5x_5 \leq 39 \\ & x_1 + 4x_2 + 2x_3 + 3x_5 \leq 64 \\ & x_1 + 2x_2 + 3x_4 + x_5 = 15 \\ & x_1, x_2, x_3, x_4, x_5 \geq 0\end{array}$$

a) (10 points) Report the optimal solution and optimal objective value

Part II - Sensitivity Analysis

A popular application of LP involves agricultural production planning. Suppose a small district has three farms and its Land Management Office wants to find an optimal planting schedule for the upcoming season. Each farm is limited by both the available irrigable land to plant on and quantity of water allocated for irrigation by the district's Water Management Department.

The crops suitable for planting in this region include sugar beets, cotton, and corn. These crops differ in their expected net return per acre and the amount of water they require. The local officials have also set a maximum quota for the total acreage that can be devoted to any one of these crops among the three farms. Additionally, in order to maintain equity among all the farms, district officials have imposed a strict requirement that each farm must plant the same proportion of its available land (e.g., if farm 1 plants 50% of its available acres, then farms 2 and 3 must also plant on 50% of their available acres).

The tables below describe the problem data.

Available acreage and water resources for each farm

Farm	Available Acres	Available Water (Acre-feet)
1	400	610
2	600	800
3	300	375

Maximum quota, water consumption, and net return for each crop

Crop	Maximum Quota (Acres)	Water Consumption (Acre-feet / Acre)	Net Return (\$/Acre)
Sugar beets	605	3	1000
Cotton	500	2	750
Corn	325	1	250

Given the above information, the Land Management Office wants to plan how many acres to devote to each crop at each farm in order to maximize the total net return, all while satisfying the requirements.

- Develop an LP model for the above problem.
- Solve the model using Excel and generate the sensitivity report to answer the questions in assignment "Optimization Sensitivity Analysis – Farm Planning Problem" on Canvas. Upload the Excel file as response to the last question. Also, solve the model using GUROBI and upload a snapshot of your screen showing the answer and date/time stamp.

